Due: August 13, 2026

1. Suppose you are asked to rate the following beers in order of preference: Budweiser, Samuel Adams, Dos Equis, Harp, Bass, and Goose Island. How many possible orderings are there?

2. Suppose you are asked to pick your three favorite beers from the list above and rank those three in order of preference. How many possible arrangements are there?

3. Suppose you are asked to identify your four favorite beers from the list above, but are not required to rank them in order of preference. How many four-beer combinations are possible?

4. Evaluate the following:

   a) 10!

   b) ${}_{10}P_9$

   c) ${}_{10}C_9$

5. Let $S = \{e_1, e_2, e_3, e_4\}$ be a sample space.

   a) Is the assignment of probabilities $P(e_1) = 0.2$, $P(e_2) = 0.4$, $P(e_3) = -0.1$, $P(e_4) = 0.5$ valid? Why or why not?

   b) Is the assignment $P(e_1) = 0.3$, $P(e_2) = 0.4$, $P(e_3) = 0.1$, $P(e_4) = 0.1$ valid? Why or why not?

   c) If $P(e_1) = 0.1$, $P(e_2) = 0.4$, and $P(e_3) = 0.3$, what must $P(e_4)$ be for this to be a valid probability distribution?

6. Suppose a researcher estimates that the probability of a country experiencing a civil war in a given year is 0.05, and the probability of experiencing a coup is 0.08. The probability of experiencing both a coup and a civil war in the same year is 0.01.

   a) What is the probability of a country experiencing either a civil war or a coup (or both)?

   b) What is the probability that a country experiences neither event?

   c) Are the events "experiencing a civil war" and "experiencing a coup" independent? Show your work.

7. In a particular state, 60% of voters are registered Democrats, 30% are Republicans, and 10% are Independents. During a recent election, 90% of Democrats, 10% of Republicans, and 50% of Independents voted for the Democratic candidate.

   a) What is the total probability that a randomly selected voter voted for the Democratic candidate? (Hint: Use the Law of Total Probability).

   b) Given that a voter voted for the Democratic candidate, what is the probability that they are a registered Republican? (Hint: Use Bayes' Rule).

8. Consider a discrete random variable X that represents the number of terms a prime minister serves. Its probability mass function (PMF) is given by:

$$P(X = x) = \begin{cases} 0.4 & \text{for } x = 1 \\ 0.3 & \text{for } x = 2 \\ 0.2 & \text{for } x = 3 \\ 0.1 & \text{for } x = 4 \\ 0 & \text{otherwise} \end{cases}$$

a) Find the Cumulative Distribution Function (CDF), $F(x)$, of X.

b) Calculate the expected value (mean) $E[X]$.